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# A polysaccharide from cultured mycelium of *Hericium erinaceus* and its anti-chronic atrophic gastritis activity

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## Abstract

A polysaccharide named EP-1 was found by screening cultured mycelium of *Hericium erinaceus*, which was extracted and subjected to precipitation with ethanol, hollow-fiber ultrafiltration and ion–exchange chromatography. The polysaccharide has a molecular weight of approximately 3100 Da and is composed of glucose, mannose and galactose, thus being a heteroglycan. EP-1 has a backbone of α-D-Glc(1→3) and β-D-Glc(1→3).The β-D-Glc(1→3) and α-D-Gal-(1→3) were regarded as branches attached to the C-4 position. The α-D-Man was regarded as a terminal residue. The anti-CAG activity was evaluated in experimental systems using a cell model for identification. The polysaccharide significantly inhibited the growth of MC cells obtained from human gastric mucosa epithelium (GES-1) cells transformed by MNNG, which were used as a chronic atrophic gastritis cell model. It also interfered with the MC cells by inducing cell cycle arrest. Thus, EP-1 shows potential for the development of new functional foods and drugs.

**Keywords:** polysaccharide; anti-CAG activity; *Hericium erinaceus*

## 1. Introduction

*Hericium erinaceus* (HE) is well known as a valuable food product and as an important medicinal fungus. It is very popular in Asia, particularly in Japan and China **[1]**. Its fruiting bodies have been used in traditional Chinese medicine and as a home remedy for chronic atrophic gastritis (CAG) and several other diseases, and thus, the fruiting bodies of this fungus have attracted considerable attention due to their

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beneficial health effects **[2]**. The fruiting body of HE contains a polysaccharide as its major component, and a series of polysaccharides from the fruiting body have been purified and their chemical structures studied **[3]**. Certain structural patterns of polysaccharides have been found in some mushrooms. There is substantial evidence that the biological activities of polysaccharides are dependent on structural features such as the sugar composition, type of glycosyl linkage, branch structure, molecular weight and concentration **[4].** In this field, β-(1→3), (1→6) glycosidic linkages are believed to play important roles in enhancing the immunomodulatory effect **[3,5].**

Because naturally originating fruiting bodies of HE are rare and expensive, the extract of cultured HE mycelium is widely used in the food and pharmaceutical industries in China as a source of food supplements and drugs from HE. The extract of the HE mycelium culture has anti-CAG activity and has been developed as a new drug named "Houtouguojun Pian" **[6]**. However, the active anti-CAG components of the extract and of "Houtouguojun Pian" have not yet been identified **[7]**. Therefore, in this study, we screened the fractions of the HE mycelium culture extract with high anti-CAG activity and identified a specific polysaccharide component as the principal active component. The physicochemical properties and chemical characteristics of the active component were further studied.

## 2. Materials and methods

### 2.1. Reagents and materials

The extract of the HE mycelium culture complies with the National Drug Standards No. HI4023098 and was obtained from Hangzhou Johncan International. An ultrafiltration system and filtration membranes were acquired from the Millipore Company Ltd. (USA), and dialysis tubes (USA, MD10), standard monosaccharides and BSA were obtained from Sigma Co. Ltd. (USA). The Wei Lexin that was used as a positive control in the pharmacological experiments was acquired from the Pharmaceutical Company of Jilin University. The other chemicals and reagents were analytical grade and obtained from local sources.

### 2.2. Cells

The anti-CAG effect of HE was evaluated using a cell model (MC cell lines). The cell lines were provided by the Clinical Oncology Hospital and Molecular Genetics and Molecular Biology Laboratory, Beijing Institute of Oncology, Peking University and maintained as a monolayer in DMEM or RPMI-1640 medium supplemented with 10% fetal calf serum, in a humidified atmosphere containing 5% $CO_2$. Cells in the



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logarithmic phase were used in the experiments. The MC cell line is a precancerous transformation of the human gastric epithelial cell line (GES-1) induced by N-methyl-N′-nitro-N-nitrosoguanidine nitroguanidine (MNNG), and it is usually used for evaluating drugs for anti-CAG activity**[8]** .

**2.3. Extraction and purification of polysaccharide**

The cultured HE mycelium was extracted with 5 volumes of water at 70°C under gentle stirring for 12 h. The residue was re-extracted twice as described above, and the supernatants were combined, concentrated and precipitated with ethanol (final concentration 80%) to obtain an ethanol-soluble fraction and an ethanol-precipitated fraction. The ethanol-precipitated fraction was further separated into two fractions by passing it through hollow fiber ultrafiltration cartridges (3 K). The resulting fractions were a concentrated solution fraction and a permeate fraction. The permeate fraction was further separated by a hollow fiber ultrafiltration column of 0.2 K to obtain the concentrated solution and permeate fractions. Then, the concentrated solution was applied to a DEAE-Sephadex column and eluted stepwise with distilled water, 0.2 M sodium chloride and 2.0 M sodium chloride. The fraction eluted with distilled water was concentrated and freeze-dried to obtain EP-1, which was then subjected to structural characterization and the determination of biological activity.

**2.4. Analysis of the chemical and physical properties of the samples.**

The total carbohydrate, uronic acid and protein contents were quantified by the phenol–sulfuric acid**[9]**, m-hydroxydiphenyl **[10]** and Bradford methods**[11]**, respectively, using glucose, glucuronic acid, and bovine serum albumin (BSA) as standards, respectively. The sugar components were analyzed by converting the sugars into PMP derivatives**[12]**, which were then detected by HPLC. HPLC was performed using a Shimadzu 2010 instrument equipped with a C-18 column and controlled by a Uniport HP N-2000 data station.

**2.5. Analysis of the purity and molecular weight**

The purity and molecular weight were determined by HPLC system( Shimadzu LC-2010A) equipped with a refractive index detector on an OH-park column equilibrated with 0.7% sodium sulfate, and calibration curves were obtained through comparison with standard dextrans and calculated with GPC software from the National Institute for the Control of Pharmaceutical and Biological Products of China.

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**2.6. Methylation Analysis**

The sample was methylated once by the Ciucanu method **[13]**. The resulting



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partially methylated alditol acetates were then analyzed by GC (Shimadzu GC-2010) and GC–MS (Shimadzu 2010) using a DB-1 capillary column (30 m × 0.25 mm) to detect the positions and quantities of glycosidic linkages in the samples. The injector temperature was 250°C, and the column temperature was held at 60°C for 5 min and then increased to 220°C at a rate of 5°C/min. The samples of partially methylated alditol acetates were dissolved in acetone and injected as 1 µL aliquots. The molar ratios of the component sugars were determined based on the peak areas and the response factors **[14]** of the flame ionization detector in the GC. Mass spectra were recorded in the positive ion electron ionization (EI) mode with MSD ChemStation.

**2.7. FT-IR spectral analysis**

FT-IR (Spectrum 100, Perkin Elmer, USA) was used to investigate the vibrations of the molecules and polar bonds among different atoms. EP-1 was mixed with KBr powder, ground thoroughly and then pressed into a 1 mm pellet for FT-IR measurement. FT-IR spectra of the samples were recorded in the frequency range of 4000 to 500 $cm^{-1}$**[15]**.

**2.8. NMR analysis**

Dried EP-1 (30 mg) was deuterium exchanged several times by lyophilizing with $D_2O$. $^1H$ NMR and $^{13}C$ NMR were recorded on a Bruker AV 600 spectrometer (Bruker, Rheinstetten Germany) at 25°C. Date processing was performed using the standard Bruker NMR software. The chemical shift was expressed in ppm.

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**2.9. Anti-CAG activity of EP-1**

**2.9.1. Inhibition of the growth of MC cells in vitro by MTT assay**

Briefly, the inhibition of the growth of MC cells was assessed using the 3-[4, 5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT) microculture tetrazolium viability assay**[16].** MC cells ($5 \times 10^4$ cells/well) in DMEM were plated onto a 96-well microplate and treated with serum containing increasing amounts of HE extract (0.5, 1, 2 and 4 mg/ml). Each experimental group consisted of six wells. After 24 h of incubation, 10 µl of MTT (5 mg/ml) was added to each well, and the plates were incubated for 4 h in the dark. Following the removal of the medium, 150 µl of DMSO was added to each well to solubilize the formazan crystals. The OD value was read at 490 nm using a microplate reader.

**2.9.2. Cell cycle analysis of EP-1-treated MC cells**

MC cells were plated onto a 6-well plate at $8\times10^5$/well and cultivated overnight. Then, 100 or 500 µg/ml of EP-1 was added to each well, and the cells were harvested



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by trypsinization for 24 h. The harvested cells were incubated with 500 µl of ice-cold 70% ethanol/well at 4°C overnight. After ethanol treatment, the cells were washed with 1 ml ice-cold PBS/well, re-suspended in 100 µl PBS/well and incubated with 300 µl propidium iodide (PI) (Sigma, Saint Louis, Missouri, USA) solution (3 µl RNase and 20 µg PI per ml)/well in the dark at 37°C for 30 minutes. The stained cells were analyzed using a FACS Calibur flow cytometer (Becton Dickinson), and 0.002% (v/v) DMSO-treated MC cells were used as the control and analyzed as described above.

**2.10. Statistical analysis**

Experiments were repeated three times. Values were shown as the mean±standard deviation (SD). The Student's t-test was performed for statistical evaluation using the SPSS 16.0 statistical software package (SPSS Inc., USA). $P < 0.05$ was set as the level of statistical significance.

**3. Results and discussion**

**3.1. Extraction and purification of crude polysaccharide**

The crude polysaccharide was isolated from screening the hot water extract of cultured mycelium of *Hericium erinaceus* and was extracted and subjected to ethanol precipitation with a yield of 21.12%. Then, it was further fractionated using a hollow fiber column with pore sizes of 3 K and 0.2 K. The fraction was then further purified by DEAE Sephadex A-50 column chromatography. The sample was applied to the column, which had been pre-equilibrated with water, and successively eluted with water, 0.2 M sodium chloride (dialyzed against with water for 3 days), and 2 M sodium chloride (dialyzed against with water for 3 days). The screening and purification procedure is shown in Figure 1. Three fractions, EP-1, EP-2, and EP-3, were obtained. The recovery of the EP-1 fraction was the highest (70.5%). The recovered sample amounts of EP-2 and EP-3 were quite low (15.2% and 8.5%, respectively). Therefore, only EP-1 was used to evaluate the anti-CAG activities. The screening and purification procedure is shown in Fig. 1.

**Fig. 1.**

**3.2. Analysis of the chemical and physical properties**

The total carbohydrate and protein contents of EP-1 were 95.7% and 3.4%. The uronic acid content of EP-1 was not detected. The sugar components were analyzed by converting the sugars into PMP derivatives. The results showed that EP-1 consisted of mannose (Man), glucose (Glc) and galactose (Gal) in a molar ratio of 6.42:67.87:1.00. High-performance gel permeation chromatography (HPGPC) was

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performed to determine the molecular weights of EP-1 using dextran standards and the HPGPC chromatogram. The fraction of EP-1 was eluted as a single symmetrical peak, suggesting that EP-1 was a homogeneous polysaccharide, and EP-1 had a molecular weight of approximately 3100 Da .

### 3.3. Methylation Analyses of EP-1

The results of the methylation analysis of EP-1 revealed that the glucosyl residues are mainly composed of (1→3)-linked Glc*p* with approximately 10% each of (1→)- man*p* units and (1→3,4)- Glc*p* units and 1.5% of (1→3,4)- gal*p* units. The ratios of each residue are summarized in Table 1.

**Table1**. Results of the methylation analysis of EP-1 by GC-MS

| Fragment | Configuration | Molar percentage (%) | RT(min) | Major mass fragment (m/z) |
|---|---|---|---|---|
| 1,5-Di-acetyl-2,3,4,6-tetra-*O*-methyl mannitol | 1→- man*p* | 11.8 | 16.478 | 71, 87, 101, 117, 145, 161, 189, 205 |
| 1,3,5-Tri-acetyl-2,4,6-tri-*O*-methyl glucitol | →3)-glu*p*-(1→ | 75.8 | 18.673 | 71, 87, 101, 117, 129, 161, 233, 277 |
| 1,3,4,5-Tetra-acetyl-2,6-di-*O*-methyl glucitol | →3,4)- glu*p* -(1→ | 10.9 | 19.837 | 71, 87, 101, 117, 129, 261, 305, 333 |
| 1,3,4,5-Tetra-acetyl-2,6-di-*O*-methyl galactitol | →3,4)- gal*p* -(1→ | 1.5 | 21.878 | 71, 87, 101, 117, 129, 261, 305, 333 |

### 3.4. Analysis of FT-IR spectra of EP-1

To more precisely characterize the polysaccharide fractions, the characteristic absorption of polysaccharide was analyzed in the region 4000-500 $cm^{-1}$ by FT-IR spectrum. As shown in **Fig. 2**, the intense broad peak at 3379 $cm^{-1}$ was characteristic of the stretching vibration of hydroxyl groups, and the weak absorption peak at 2929 $cm^{-1}$ was ascribed to the stretching vibration of the C-H bond. In addition, the asymmetrical stretching peak near 1412 $cm^{-1}$ for EP-1 indicated the presence of carboxyl groups, being attributed to the bending vibration of C-H or C-O **[17]**. Each particular polysaccharide has a specific bond in the region of 1200-1000 $cm^{-1}$.This region is often considered as the finger print of a molecule because it allows the identification of major chemical groups in polysaccharides based on the position and intensity of bond peaks that are specific to each polysaccharide **[18]**. The peaks located at 1154, 1080 and 1025 $cm^{-1}$ corresponded to the stretching vibrations of (C-OH) side groups and the (C-O-C) glycosidic bond vibration, indicating a pyranose unit in each polysaccharide. The absorptions at 861 and 762 $cm^{-1}$ indicated that EP-1 contained both β-glycosidic and α-glycosidic linkages between the sugar units **[19]**.

**Fig. 2.**

### 3.5.Determination of the sugar residues and their sequence by NMR

The structural properties of EP-1 were further elucidated by NMR spectroscopy.



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The 600-MHz spectra are shown in Fig. 3. The $^{1}$H-NMR spectrum of EP-1 consisted mostly of signals for five anomeric protons at δ 5.335,5.167,4.597 and 4.584 ppm (Fig. 3A). The chemical shifts at 5.335 and 5.167 ppm were previously assigned to an α-glycosidic linkage, and the signals at δ 4.597 and 4.584 ppm were assigned to a β-glycosidic linkage**[20]**. These findings were consistent with our FT-IR spectrum, which indicated that EP-1 has both α- and β-configurations.

The $^{13}$C-NMR spectrum of EP-1 displayed four strong signals in the region from δ 110 to 90 ppm. The peaks from δ 100 to 99 ppm were ascribed to α-D-Man(1→ and →1)-α-D-Glc (3→. The chemical shifts at δ 95.61 and 95.49 ppm were assigned, respectively, to →1)-β-D--Glc(→3,4 and →1)-α-D-Gal-(→3,4 **[21]**. The signals at 26.96 ppm were assigned to the methyl groups of Gal **[22]**. Taken together, these findings indicate that EP-1 has a backbone of α-D-Glc(1→3) and β-D-Glc(1→3) .The β-D--Glc(1→3 and α-D-Gal-(1→3 were regarded as the branches attached to the C-4 position, and α-D-Man was regarded as a terminal residue.

**Fig. 3.**

### 3.5. Anti-CAG activity of EP-1

#### 3.5.1 Inhibitory effects of EP-1 on MC cells

Different doses of EP-1 were tested by the MTT method, and the results are shown in **Fig. 4**. The results show that EP-1 exhibited an inhibitory effect on the proliferation of MC cells, and the inhibitory effect was dose dependent from 0.5 to 4 mg/ml.

**Fig. 4.**

All of the diagrams in Fig. 4 represent the typical results of inhibitory effects experiments. All of the data are expressed as the mean±SD of six wells performed in triplicate. * and ** denote P< 0.05 and P< 0.01, respectively, compared to untreated cells using Student's t-test.

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#### 3.5.2 EP-1 interfered with MC cell proliferation by inducing cell cycle arrest

According to the above results, EP-1 could exert an inhibitory impact on MC proliferation. To discover which mechanism was involved in the inhibition of MC cell proliferation, EP-1-exposed MC cells were first investigated by cell cycle analysis using flow cytometry. Cell cycle analyses conducted after the incubation of MC cells with a series of increasing concentrations for various durations demonstrated that EP-1 arrested MC cells in the G0/G1 phase in a dose-dependent response (**Fig. 5**). In addition, EP-1 treatment increased S population compared to the untreated MC cells. However, G2/M arrest induced by EP-1 was not statistically significant.



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**Fig. 5.**

(A) Normal control group; (B) incubated with 100 μg/ml of EP-1; (C) incubated with 500 μg/ml of EP-1 and (D) mean±SD of three independent experiments.

* and ** denote $P < 0.05$ and $P < 0.01$, respectively, compared to the normal control group using Student's t-test.

### 3.6. Discussion

It has been reported elsewhere that the extract of cultured HE mycelium contains many components, such as amino acids, sugars, proteins, polysaccharides, erinacine, erinapyrone, and sterols **[23]**, and has anti-CAG activity, but the precise identity of the principal active component still remained to be clarified. In this study, we performed a primary fractionation of the aqueous extract of the cultured HE mycelium by ethanol precipitation and found that the anti-CAG activity separated into the ethanol-precipitated fraction. This result indicated that the active component is likely to be a polysaccharide rather than a water-soluble small organic compound such as an amino acid. Further purification of the ethanol-precipitated fraction by molecular size exclusion and by DEAE column chromatography identified one specific fraction, EP-1, as containing the anti-CAG components of the cultured HE mycelium extract.

The anti-CAG activity of the EP-1 fraction came from a polysaccharide with a molecular weight of approximately 3100 Da, and it was revealed that the molecular weight of the polysaccharide has an important influence on the anti-CAG effect. Although the active crude polysaccharide fractions obtained from the purification steps consisted of glucose, mannose, galactose and uronic acid, the purified polysaccharide did not contain uronic acid, which suggests that uronic acid is not necessary for the observed activity. The main chain of EP-1 was mainly composed of α-D-Glc(1→3) and β-D-Glc(1→3), and some side chains were linked to the main chain through the 4 position of the main chain and were terminated by α-D-man*p*. The (1→3)-linked Glc*p* was predominant in the heteroglycan, which suggests that the (1→3)-linked glucan structure might play an important role in its anti-CAG activity.The possible structure of EP-1 was established:

$$\alpha\text{-D-Man}p\text{-}(1\rightarrow 3)\text{-}\alpha\text{-D-Glc}p\text{-}(1\rightarrow 3)\text{-}\left(\alpha\text{-D-Glc}p\text{-}(1\rightarrow 3)\right)_n\alpha\text{-D-Glc}p\text{-}(1\rightarrow$$

$$\begin{array}{c} 4 \\ \uparrow \\ 1 \\ \alpha\text{-D-Man}p\text{-}(1\rightarrow 3)\text{-}\beta\text{-D-Glc}p\,(\text{or Gal}p^{*}) \end{array}$$

*means only approximately 10% Gal*p* was linked with the main chain.

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The results of this study showed that EP-1 interfered with MC cell proliferation by inducing cell cycle arrest. However, the exact anti-CAG mechanism is not known. The observed anti-CAG activity may be due to both the cell cycle arrest and apoptosis. Further detailed study of various other parameters could illuminate other mechanisms of action.

## 4. Conclusions

In the present study,we reported a new anti-CAG polysaccharide EP-1 isolated from HE mycelia culture.The structural feature is different from other HE polysaccharides published previously. EP-1 interfered with MC cell proliferation by inducing cell cycle arrest, suggesting this polysaccharide has good ability to anti-CAG.


## Acknowledgments

This work was supported by the National Natural Science Foundation of China [Grant No. 308773370]. It was also supported by the Foundation of Science and Technology Department in Jilin Province [Grant No. 20060559]. The authors thank Professor Xu Xin (Changchun University of Chinese Medicine) and Professor Tetsuya Konishi (Niigata Univ. Pharm & Appl. Life Sci.) for their valuable suggestions and assistance in the experiments.


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**Table 1**. Results of the methylation analyses of EP-1 by GC-MS

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**Fig. 1.** Summarized extraction and isolation procedure

**Fig. 2.** FT-IR spectrum of EP-1

**Fig. 3.** NMR spectra of EP-1. (A) $^{1}$H-NMR and (B) $^{13}$C-NMR

**Fig. 4.** MC cells survival measured by the MTT assay

All of the diagrams in Fig. 3 are the typical results of three independent experiments. All of the data are expressed as the mean±SD of six wells performed in triplicate. * and ** denote P< 0.05 and P< 0.01, respectively, compared to untreated cells using Student's t-test.

**Fig. 5.** Induction of MC cell cycle by EP-1

(A) The normal control group; (B) incubated with 100 µg/ml of EP-1; (C) incubated with 500 µg/ml of EP-1 and (D) mean±SD of three independent experiments.

* and ** denote P< 0.05 and P< 0.01, respectively, compared to the normal control group using Student's t-test.

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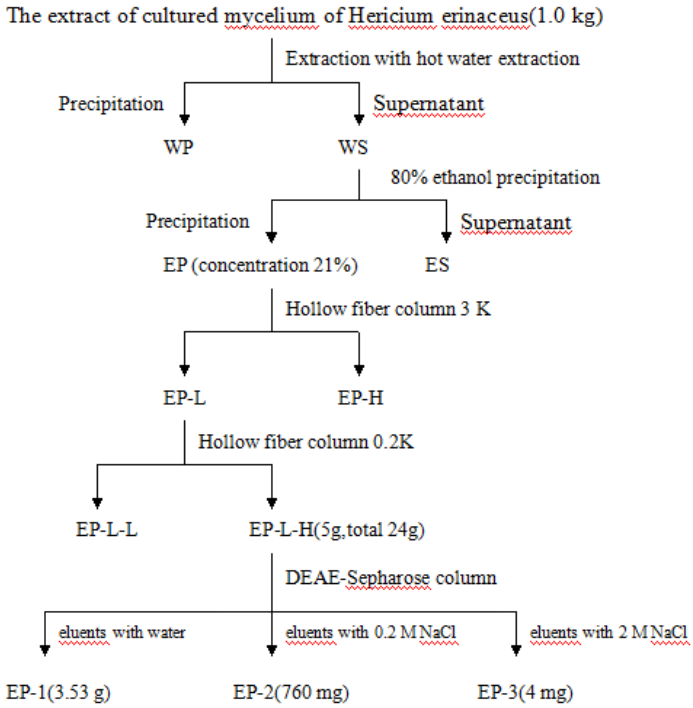


**Fig. 1.Summarized extraction and isolation procedure**

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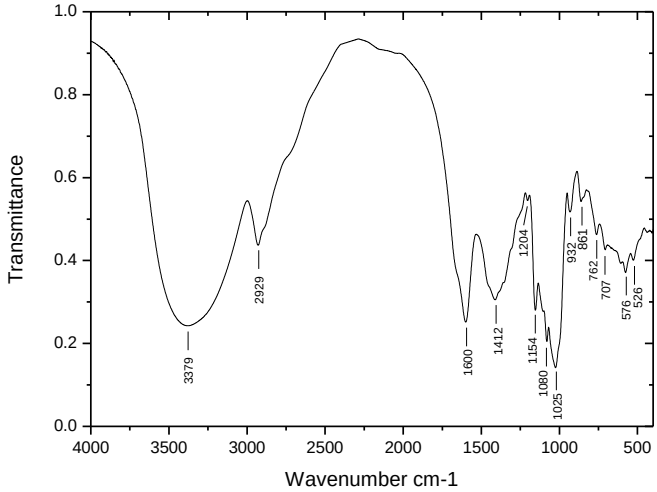


**Fig. 2. FT-IR spectrum of EP-1**

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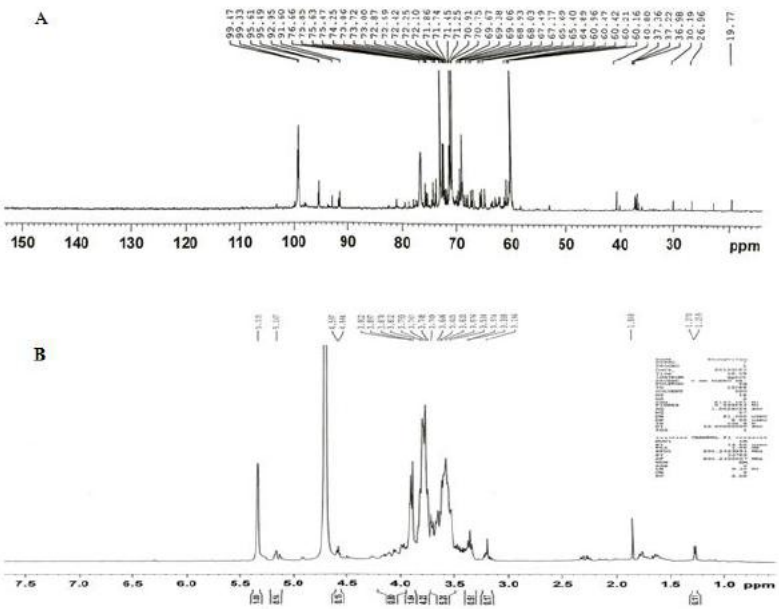


**Fig. 3. NMR spectra of EP-1. (A) $^{1}$H-NMR and (B) $^{13}$C-NMR**

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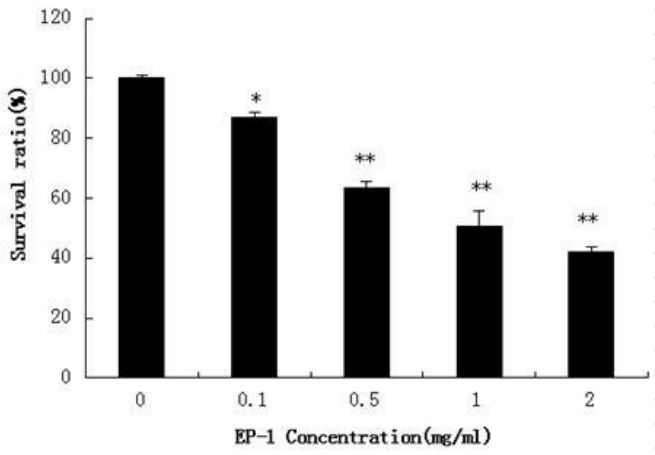


**Fig. 4. MC cells survial measured by the MTT assay**

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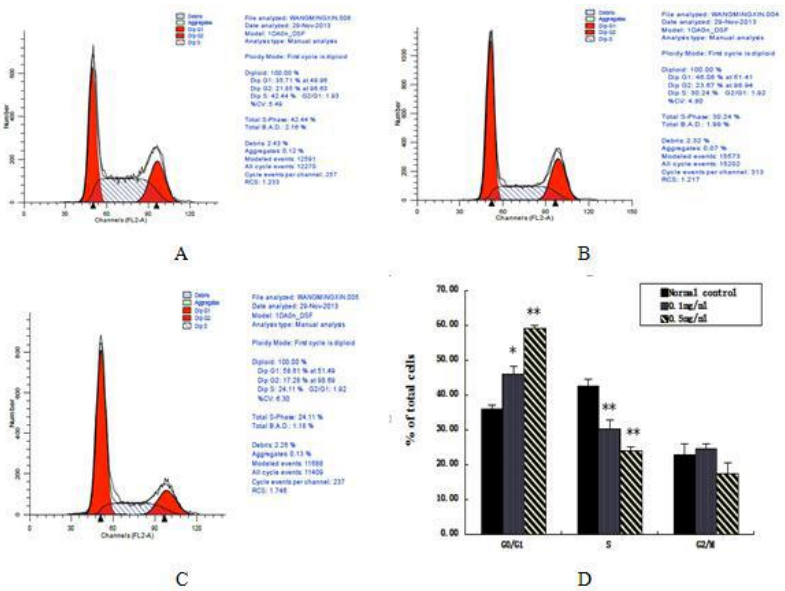


**Fig. 5. Induction of MC cell cycle by EP-1**

(A) The normal control group; (B) Incubated with 100 µg/ml of EP-1; (C) Incubated with 500 µg/ml of EP-1 and (D) mean±SD of three independent experiments.

* and ** denoted P< 0.05 and P< 0.01 respectively as compared to the normal control group using Student's t-test.



Accepted